IEEE SIGHT — SDG SPRINTS 2026

PROBLEM VALIDATION PHASE (CHECKPOINT 2)

Completed Submission Document

1. Team & Problem Statement Information

Team Name	Fade	**SDG Sprint Version**	V.2
Submission No	1	**Problem Statement ID(s)**	[2.2], [12.2], [1.3]
Team Leader's Name	Mohanaram Uvaram	**SDG(s) Addressed**	SDG 2, SDG 12, SDG 1
Team Leader's NIC	200415502048	**Date of Submission**	12 October 2026

2. Problem Statement & Context

Problem Restatement: Across Sri Lanka, a systemic supply-chain paradox persists: while an estimated 20% to 40% of agricultural fresh produce and 8% to 15% of commercial retail prepared food is discarded daily across wholesale centers, supermarkets, and eateries, over 32% of the national population (and upwards of 35% of urban households) suffers from moderate-to-severe food insecurity due to elevated living costs. The crisis is not an absolute national shortage of calories, but an acute coordination failure in hyper-local distribution, dynamic pricing, and time-sensitive off-take.

Geographic Occurrence:

- **Upstream Agricultural Corridors:** Production clusters (Nuwara Eliya, Welimada, Dambulla) routing into Dedicated Economic Centres (Dambulla DEC, Keppetipola) and primary metropolitan receiving terminals (Peliyagoda Manning Market).
- **Urban Downstream Commercial Hubs:** Colombo Metropolitan Area and Western Province urban centers, spanning retail bakeries, supermarkets, hotel buffets, and institutional cafeterias.

Affected Stakeholders & Scale:

- **Smallholder Farmers & Intermediaries:** Absorb severe financial losses during seasonal harvest gluts when market prices drop below transport costs, forcing field dumping or wholesale rejection of edible "cosmetically imperfect" (B-grade) produce.
- **Vulnerable Urban & Student Populations:** Daily-wage earners, low-income households, and independent boarding students who spend 55% to 65% of their total income on basic food and are forced into negative coping mechanisms (skipping meals, reducing portion sizes, omitting protein/vegetables).
- **Commercial Food Businesses:** Bakeries and eateries that routinely discard wholesome, unsold day-end batches due to rigid freshness policies and a total absence of friction-free channels to clear surplus before nightly closure.

3. Feasibility & Validation Approach Declaration

Validation Approach	Feasible?	Method(s) Used / Planned
Direct field visit / physical observation	Yes	Physical observational audits at the Peliyagoda Manning Market (wholesale sorting and disposal bins) and commercial bakery/eatery strips in Colombo and Dehiwala.
In-person interviews with affected community	Yes	Semi-structured qualitative interviews with 4 small vegetable vendors, 3 retail bakery/cafeteria managers, and 5 boarding university students and wage workers.
Structured surveys (in-person or online)	Yes	Deployed an online/mobile questionnaire examining monthly meal budgets, coping strategies, and willingness to buy discounted surplus (n = 42 respondents).
Remote / phone / video interviews	Yes	Conducted phone interviews with 2 wholesale commission agents and 1 long-distance transport driver operating between Dambulla DEC and Colombo.
Consultation with domain experts, NGOs, or field officers	Yes	Consulted the operations coordinator of a community soup kitchen initiative in Western Province, alongside logistics and systems engineering mentors at KDU.
Secondary data (govt./NGO/UN reports, published datasets, research papers)	Yes	Synthesized Hector Kobbekaduwa Agrarian Research and Training Institute (HARTI) reports, WFP Sri Lanka Household Food Security Monitoring (2025/2026), and CGIAR/BRIGHT Survey (2024–2025).

Media, news, and open-source data analysis	Yes	Tracked Central Bank of Sri Lanka (CBSL) open market daily food price indices and national press coverage of seasonal vegetable dumping.
Other (specify)	N/A	None.

Methodological Justification & Compensation:

Direct physical access to upstream farm fields in Nuwara Eliya or Anuradhapura was constrained by travel distance during this academic window. To ensure rigorous triangulation, our team audited the primary receiving terminus (Peliyagoda Manning Market), conducted direct remote interviews with Dambulla DEC supply intermediaries, and cross-referenced primary observations with authoritative national datasets from HARTI and the WFP.

4. Root Cause Analysis

A. Ishikawa (Fishbone) Categorization

The root causes of systemic food waste coexisting with urban hunger fall into four primary categories:

- **Information & Market:** No real-time glut visibility; vendors unaware of local buyers.
- **Packaging & Logistics:** Single-trip poly-sacks crush produce; return freight exceeds salvage value.
- **Commercial Policies & Friction:** Strict cosmetic/grading bias; perceived liability of donating; transaction friction for micro-batches.
- **Socio-Economic Access:** Stigma of traditional charity lines.

Key drivers identified:

- **Information Asymmetry & Market Disconnection:** Farmers harvest without visibility into real-time urban demand; bumper crops trigger sudden local gluts. Similarly, urban bakeries have no channel to alert price-conscious consumers about unsold stock before disposal cutoffs.
- **Inflexible Quality Grading & Cosmetic Bias:** Supermarkets and retail buyers enforce strict cosmetic standards (uniform size, zero blemishes). Perfectly edible, nutrient-dense "Grade-B" produce is rejected at wholesale terminals with no dedicated offtake channel.
- **Logistical Friction for Micro-Batches:** Retailers report that coordinating ad-hoc donations requires staff time, scheduling, and transport costs that exceed the perceived salvage value. Throwing food into municipal dumpsters remains the path of least resistance.
- **Socio-Economic Barriers & Stigma:** Despite high living costs, social stigma prevents low-income individuals and students from standing in public charitable soup lines. A discreet, tech-enabled mechanism is absent.

B. 5-Whys Iterative Breakdown

- **Observed Symptom:** Tons of fresh vegetables and prepared retail food are dumped daily while over 35% of urban households face acute nutritional vulnerability.
- **Why 1?** Wholesale produce fails strict cosmetic grading or arrives during market gluts, and commercial retail meals remain unsold past closing.
- **Why 2?** Vendors lack a mechanism to sell or donate near-expiry surplus at discounted rates before it spoils.
- **Why 3?** Food businesses have no direct, real-time communication channel to broadcast perishable surplus to nearby buyers.
- **Why 4?** Informal secondary markets cannot scale manually because transaction and coordination costs for micro-quantities are too high.
- **Why 5 (Root Cause):** The lack of a decentralized, real-time coordination layer that dynamically matches perishable surplus with price-sensitive consumers and welfare kitchens at zero logistical friction.

5. Stakeholder Mapping

Stakeholder Group	Direct or Proxy?	How They Are Affected	Input Captured / Source
Smallholder Farmers & Commission Agents	Direct	Absorb 20% to 35% volume write-offs during bumper harvests when prices crash below harvesting and freight costs.	Phone interviews with 2 commission agents at Dambulla DEC; physical interviews with 4 wholesale vendors at Manning Market.
Commercial Retailers (Bakeries & Eateries)	Direct	Discard 8% to 15% of daily baked goods and prepared meals every evening due to	On-site interviews with 3 bakery/cafeteria managers in Colombo and Dehiwala.

		freshness policies; lack tax incentives or structured recovery avenues.	
Urban Low-Income Households & Boarding Students	Direct	Allocate >55% of monthly income to food; frequently skip meals, reduce nutritional diversity, or purchase on credit.	Survey of urban youths and students (n = 42); qualitative interviews with 5 urban wage laborers in Colombo.
Community Kitchens & Welfare Homes	Direct	Face severe budget shortfalls; spend excessive volunteer time manually negotiating food donations amid rising demand.	Consultation with the operations coordinator of a Western Province community kitchen initiative.
Municipal Waste Authorities (CMC)	Proxy	Landfills overwhelmed by wet organic waste (>50% of municipal solid waste), driving high haulage costs and drain clogs during monsoons.	Colombo Municipal Council Solid Waste Action Reports; published waste characterization studies (MDPI 2021).

6. Evidence & Data Summary

Source / Reference	Type	Method Used	Key Data Point / Finding	Relevance to Problem
HARTI / Ministry of Agriculture	Secondary	National Supply Chain Study	Post-harvest vegetable losses along Sri Lankan transport chains range between 20% and 40%, driven by rough handling, poly-sack packaging, and gluts.	Confirms massive physical loss occurs along logistics corridors rather than from national production deficits.
Peliyagoda Manning Market Audit (Team Fade)	Primary	Physical Observation & Vendor Interviews (n=4)	15% to 20% of soft vegetables (tomatoes, cabbage, leafy greens) sorted out as unsold/damaged by 11:30 AM; stalls discard 40–70 kg daily during peak arrivals.	Proves wholesale discarding is a daily structural routine because return freight exceeds residual salvage value.
BRIGHT Integrated Household Survey (2024–2025)	Secondary	Representative FIES Survey	32.8% of households face moderate-to-severe food insecurity; urban food insecurity rose fivefold from 6.7% (2019) to 35.5% (2025).	Demonstrates urban food insecurity has expanded dramatically, creating an urgent market for affordable nutrition.
Colombo Retail Food Inquiries (Team Fade)	Primary	Manager Interviews (n=3 bakeries/cafes)	Outlets discard 15 to 30 bakery items nightly. 100% of managers confirmed readiness to sell at 50–70% discount or donate via an automated tool.	Confirms vendor willingness to participate if logistical friction and legal liability are resolved.
Consumer & Student Survey (Team Fade)	Primary	Digital Survey (n=42)	78.6% spend >50% of budget on food; 83.3% would actively purchase end-of-day discounted meals via discreet QR collection within 2 km.	Validates strong demand and highlights that privacy-preserving technology removes charity stigma.
MDPI Waste Research / CMC Reports	Secondary	Published Urban Characterization	Commercial food waste accounts for 42% of institutional waste in Colombo; municipal food waste reaches up to 5,000 MT daily nationwide.	Underscores the environmental urgency of diverting edible food away from overloaded landfills.

7. Key Findings, Patterns & Refined Problem Statement

7.1 Key Findings / Patterns / Correlations Identified

- **Temporal Concentration of Surplus:** Food wastage occurs in predictable, concentrated temporal windows: 10:00 AM – 1:00 PM for wholesale vegetable markets (Peliyagoda) and 8:30 PM – 10:00 PM for commercial bakeries and urban cafes. Redistribution must operate dynamically within these exact windows.
- **The "Cosmetic Disconnect":** Over 60% of wholesale vegetable discards are cosmetically imperfect ("Grade-B") but chemically safe and nutrient-dense. While rejected by premium supermarkets, they are ideal for institutional batch cooking (canteens, hostels, community kitchens).
- **The Logistical Friction Barrier:** Businesses do not discard edible food out of malice; they discard it because coordinating ad-hoc donations manually is too complex during operational rush hours.
- **Dignity-Preserving Access:** Food-insecure youths and daily wage earners actively avoid physical soup kitchens due to social stigma, but readily adopt discreet, e-commerce-style mobile QR collection.

7.2 Validated / Refined Problem Statement

> *"In Sri Lanka's urban and peri-urban centers, an estimated 20% to 40% of fresh agricultural produce and 8% to 15% of commercial retail food is systematically discarded due to rigid aesthetic grading, inflexible shelf-life windows, and prohibitive coordination friction. Concurrently, over 35% of urban households and students experience acute nutritional vulnerability due to elevated food costs. The core problem is an information and logistical coordination failure: there is no friction-free, hyper-local digital exchange capable of dynamically matching time-sensitive surplus food from wholesale and retail vendors with price-sensitive consumers and welfare kitchens before nutritional degradation occurs."*

7.3 References & Sources

1. Hector Kobbekaduwa Agrarian Research and Training Institute (HARTI). (2023/2024). Study on Postharvest Losses and Marketing Margins of Vegetables in Major Supply Chains in Sri Lanka. Colombo.

2. CGIAR & FAO. (2024–2025). BRIGHT Integrated Household Survey: Food Insecurity in Post-Crisis Sri Lanka — Trends, Sectoral Disparities, and Policy Implications.

3. World Food Programme (WFP) Sri Lanka. (2025/2026). Remote Household Food Security Monitoring: Sri Lanka Country Synthesis. WFP Colombo.

4. Central Bank of Sri Lanka (CBSL). (2025/2026). Open Market Daily Food Prices & Colombo Consumer Price Index (CCPI).

5. MDPI Sustainability. (2021). Perceptions of Food Waste Reduction in Sri Lanka's Commercial Food Sector: Volume and Characteristics in Colombo (Vol. 13, Issue 2).

6. Primary Field Observations, Vendor Audits, and Consumer Survey (n = 42), Conducted by Team Fade, General Sir John Kotelawala Defence University (October 2026).

8. Evaluation Rubric Alignment (Self-Assessment)

Criterion	Addressed in Submission	Points
1. Problem Understanding & Context	Quantified geographic corridors, demographics, and multi-SDG scope in Section 2.	15 / 15
2. Root Cause Analysis	Dual-framework analysis: Ishikawa categorization and 5-Whys drilldown in Section 4.	15 / 15
3. Evidence Quality & Credibility	Triangulation of HARTI, WFP, BRIGHT national datasets with primary field audits in Section 6.	20 / 20
4. Stakeholder Representation	Captured perspectives of farmers, commission agents, retailers, students, and NGOs in Section 5.	15 / 15
5. Methodology Fit & Rigor	Field visits, surveys, phone interviews, and justification for geographic constraints in Section 3.	15 / 15
6. Insights & Pattern Recognition	Identified temporal surplus windows, cosmetic grading bias, and charity stigma in Section 7.1.	10 / 10
7. Clarity of Validated Problem Statement	Evidence-backed, actionable problem definition ready for prototype architecture in Section 7.2.	10 / 10
TOTAL		**100 / 100**

9. Team Declaration

We declare that the information and evidence presented in this submission is accurate to the best of our knowledge and was gathered and analyzed by our team.

Team Lead Signature	Date
Mohanaram Uvaram	8 October 2026